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|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1(a)(i)   | <b>Q</b> – <u>receptor</u> ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>1</b> |
| 1(a)(ii)  | <b>R</b> – proton / hydrogen ion / $H^+$ , pump / carrier ;<br><b>S</b> – calcium (ion) / $Ca^{2+}$ , channel (protein) ;                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>2</b> |
| 1(a)(iii) | <b>T</b> – cell surface membrane / plasma membrane / plasmalemma ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>1</b> |
| 1(b)      | <i>any five from:</i><br><b>1</b> ABA / abscisic acid, binds / attaches, to receptor ;<br><b>2</b> calcium ions enter cell ;<br><b>3</b> ( $Ca^{2+}$ is) second messenger ;<br><b>4</b> proton / hydrogen ion / $H^+$ , exit, stops / is inhibited ;<br><b>5</b> potassium ions / $K^+$ , leave (cell) ;<br><b>6</b> water potential / $\psi$ , increases (in, cell / cytoplasm) ;<br><b>7</b> water leaves (cell), by osmosis OR down, $\psi$ / osmotic, gradient ;<br><b>8</b> cell, volume decreases / becomes flaccid / loses turgidity / loses turgor ; | <b>5</b> |